[Lecture Notebook] Input, Processing, and Output

MGS3001 Python Programming for Business, Spring 2025

Chad (Chungil Chae)

Fri, 21 March 2025

code_02_01_output.py

```
print('Kate Austen')
print('123 Full Circle Drive')
print('Asheville, NC 28899')
```

```
Kate Austen
123 Full Circle Drive
Asheville, NC 28899
```

code_02_02_double_quotes.py

```
# code_02_02_double_quotes.py
print("Kate Austen")
print("123 Full Circle Drive")
print("Asheville, NC 28899")
```

```
Kate Austen
123 Full Circle Drive
Asheville, NC 28899
```

code_02_03_apostrophe.py

```python
# code_02_03_apostrophe.py
print("Don't fear!")
print("I'm here!")
```

```
Don't fear!
I'm here!
```

code_02_04_display_quote.py

```python
# code_02_04_display_quote.py
print('Your assignement is to read "Hamlet" by tomorrow.')
print("""I'm reading "Hamlet" tonight.""")
print("""One Two Three""")
```

```
Your assignement is to read "Hamlet" by tomorrow.
I'm reading "Hamlet" tonight.
One Two Three
```

code_02_05_comment1.py

```python
# code_02_05_comment1.py
# This program displays a person's
# name and address.
print('Kate Austen')
print('123 Full Circle Drive')
print('Asheville, NC 28899')
```

```
Kate Austen
123 Full Circle Drive
Asheville, NC 28899
```

code_02_06_comment2.py


```python
# code_02_06_comment2.py
print('Kate Austen') # Display the name.
print('123 Full Circle Drive') # Display the address.
print('Asheville, NC 28899') # Display the city, state, and ZIP.
```

```
Kate Austen
123 Full Circle Drive
Asheville, NC 28899
```

code_02_07_variable_demo.py

```python
# code_02_07_variable_demo.py
# This program demonstrates a variable.
room = 503
print('I am staying in room number')
print(room)
```

```
I am staying in room number
503
```

code_02_08_variable_demo2.py

```python
# code_02_08_variable_demo2.py
# Create two variables: top_speed and distance.
top_speed = 160
distance = 300

# Display the values referenced by the variables.
print('The top speed is')
print(top_speed)
print('The distance traveled is')
print(distance)
```

```
The top speed is
160
The distance traveled is
300
```

36 code_02_09_variable_demo3.py

```
# code_02_09_variable_demo3.py
# This program demonstrates a variable.
room = 503
print('I am staying in room number', room)
```

37 I am staying in room number 503

38 code_02_10_variable_demo4.py

```
# code_02_10_variable_demo4.py
# This program demonstrates variable reassignment.
# Assign a value to the dollars variable.
dollars = 2.75
print('I have', dollars, 'in my account.')

# Reassign dollars so it references
# a different value.
dollars = 99.95
print('But now I have', dollars, 'in my account!')
```

39 I have 2.75 in my account.
40 But now I have 99.95 in my account!

41 code_02_11_string_variable.py

```
# code_02_11_string_variable.py
# Create variables to reference two strings.
first_name = 'Kathryn'
last_name = 'Marino'

# Display the values referenced by the variables.
print(first_name, last_name)
```

42 Kathryn Marino

43 code_02_12_string_input.py

```
# code_02_12_string_input.py
# Input
first_name = input('Enter your first name: ') # Get the user's first name.
last_name = input('Enter your last name: ') # Get the user's last name.

# Display
print('Hello', first_name, last_name) # Print a greeting to the user.
```

44 code_02_13_input.py

```
# code_02_13_input.py
# Get the user's name, age, and income.
name = input('What is your name? ')
age = int(input('What is your age? '))
income = float(input('What is your income? '))

# Display the data.
print('Here is the data you entered:')
print('Name:', name)
print('Age:', age)
print('Income:', income)
```

45 code_02_14_simple_math.py

```
# code_02_14_simple_math.py
# Assign a value to the salary variable.
salary = 2500.0

# Assign a value to the bonus variable.
bonus = 1200.0

# Calculate the total pay by adding salary
# and bonus. Assign the result to pay.
pay = salary + bonus

# Display the pay.
print('Your pay is', pay)
```


46 Your pay is 3700.0

47 code_02_15_sale_price.py

```
# code_02_15_sale_price.py
# This program gets an item's original price and
# calculates its sale price, with a 20% discount.

# Get the item's original price.
original_price = float(input("Enter the item's original price: "))

# Calculate the amount of the discount.
discount = original_price * 0.25

# Calculate the sale price.
sale_price = original_price - discount

# Display the sale price.
print('The original price is', original_price, 'Chinese Yuan')
print('The sale price is', sale_price, 'Chinese Yuan')
print('You can save', discount, 'Chinese Yuan')
```

48 code_02_16_test_score_average.py

```
# code_02_16_test_score_average.py
# Get three test scores and assign them to the
# test1, test2, and test3 variables.
test1 = float(input('Enter the first test score: '))
test2 = float(input('Enter the second test score: '))
test3 = float(input('Enter the third test score: '))

# Calculate the average of the three scores
# and assign the result to the average variable.
average = (test1 + test2 + test3) / 3.0

# Display the average.
print('The average score is', average)
```

49 code_02_17_time_converter.py

```
# code_02_17_time_converter.py
# Get a number of seconds from the user.
total_seconds = float(input('Enter a number of seconds: '))

# Get the number of hours.
hours = total_seconds // 3600

# Get the number of remaining minutes.
minutes = (total_seconds // 60) % 60

# Get the number of remaining seconds.
seconds = total_seconds % 60

# Display the results.
print('Here is the time in hours, minutes, and seconds:')
print('Hours:', hours)
print('Minutes:', minutes)
print('Seconds:', seconds)
```

50 code_02_18_future_value.py

```
# code_02_18_future_value.py
# Get the desired future value.
future_value = float(input('Enter the desired future value: '))

# Get the annual interest rate.
rate = float(input('Enter the annual interest rate: '))

# Get the number of years that the money will appreciate.
years = int(input('Enter the number of years the money will grow: '))

# Calculate the amount needed to deposit.
present_value = future_value / (1.0 + rate)**years

# Display the amount needed to deposit.
print('You will need to deposit this amount:', present_value)
```

51 code_02_19_no_formatting.py

```python
# code_02_19_no_formatting.py
# This program demonstrates how a floating-point
# number is displayed with no formatting.
amount_due = 5000.0
monthly_payment = amount_due / 12.0
print('The monthly payment is', monthly_payment)
```

52 The monthly payment is 416.6666666666667

53 code_02_20_formatting.py

```python
# code_02_20_formatting.py
# An f-string is a special type of string literal that is prefixed with the letter f
print(f'Hello world')

# F-strings support placeholders for variables
name = 'Chad'
print(f'Hello {name}.')

# Placeholders can also be expressions that are evaluated
print(f'The value is {10 + 2}.')

val = 10
print(f'The value is {val + 2}.')

# Format specifiers can be used with placeholders
num = 123.456789
print(f'{num:.3f}')
print(f'{num:.2f}')
print(f'{num:.1f}')
print(f'{num:.0f}')

num = 1000000.00
print(f'{num:,.2f}')

discount = 0.5
print(f'{discount:.0%}')

num = 123456789
```



```python
print(f'{num:,d}')

num = 12345.6789
print(f'{num:.2e}')

num = 12345.6789
print(f'The number is {num:12,.2f}')

## Alignment
print(f'{num:<20.2f}')
print(f'{num:>20.2f}')
print(f'{num:^20.2f}')

#[alignment][width][,][.precision][type]
print(f'{num:^10,.2f}')

# This program demonstrates how a floating-point
# number can be formatted.
amount_due = 5000.0
monthly_payment = amount_due / 12
print('The monthly payment is',
format(monthly_payment, '.2f'))
```

```
Hello world
Hello Chad.
The value is 12.
The value is 12.
123.457
123.46
123.5
123
1,000,000.00
50%
123,456,789
1.23e+04
The number is    12,345.68
12345.68
            12345.68
      12345.68
12,345.68
The monthly payment is 416.67
```


72 code_02_21_dollar_display.py

```
# code_02_21_dollar_display.py
# This program demonstrates how a floating-point
# number can be displayed as currency.
monthly_pay = 5000.0
annual_pay = monthly_pay * 12
print('Your annual pay is $', format(annual_pay, ',.2f'), sep='')
```

73 Your annual pay is $60,000.00

74 code_02_22_columns.py

```
# code_02_22_columns.py
# This program displays the following
# floating-point numbers in a column
# with their decimal points aligned.
num1 = 127.899
num2 = 3465.148
num3 = 3.776
num4 = 264.821
num5 = 88.081
num6 = 799.999

# Display each number in a field of 7 spaces
# with 2 decimal places.
print(format(num1, '7.2f'))
print(format(num2, '7.2f'))
print(format(num3, '7.2f'))
print(format(num4, '7.2f'))
print(format(num5, '7.2f'))
print(format(num6, '7.2f'))
```

```
75  127.90
76 3465.15
77    3.78
78  264.82
79   88.08
80  800.00
```


81 code_02_23_orion.py

```python
# code_02_23_orion.py
# This program draws the stars of the Orion constellation,
# the names of the stars, and the constellation lines.
import turtle

# Set the window size.
turtle.setup(500, 600)

# Setup the turtle.
turtle.penup()
turtle.hideturtle()

# Create named constants for the star coordinates.
LEFT_SHOULDER_X = -70
LEFT_SHOULDER_Y = 200

RIGHT_SHOULDER_X = 80
RIGHT_SHOULDER_Y = 180

LEFT_BELTSTAR_X = -40
LEFT_BELTSTAR_Y = -20

MIDDLE_BELTSTAR_X = 0
MIDDLE_BELTSTAR_Y = 0

RIGHT_BELTSTAR_X = 40
RIGHT_BELTSTAR_Y = 20

LEFT_KNEE_X = -90
LEFT_KNEE_Y = -180

RIGHT_KNEE_X = 120
RIGHT_KNEE_Y = -140

# Draw the stars.
turtle.goto(LEFT_SHOULDER_X, LEFT_SHOULDER_Y)
turtle.dot()
turtle.goto(RIGHT_SHOULDER_X, RIGHT_SHOULDER_Y)
turtle.dot()
turtle.goto(LEFT_BELTSTAR_X, LEFT_BELTSTAR_Y)
turtle.dot()
turtle.goto(MIDDLE_BELTSTAR_X, MIDDLE_BELTSTAR_Y)
```

```
turtle.dot()
turtle.goto(RIGHT_BELTSTAR_X, RIGHT_BELTSTAR_Y)
turtle.dot()
turtle.goto(LEFT_KNEE_X, LEFT_KNEE_Y)
turtle.dot()
turtle.goto(RIGHT_KNEE_X, RIGHT_KNEE_Y)
turtle.dot()

# Display the star names
turtle.goto(LEFT_SHOULDER_X, LEFT_SHOULDER_Y)
turtle.write('Betegeuse')
turtle.goto(RIGHT_SHOULDER_X, RIGHT_SHOULDER_Y)
turtle.write('Meissa')
turtle.goto(LEFT_BELTSTAR_X, LEFT_BELTSTAR_Y)
turtle.write('Alnitak')
turtle.goto(MIDDLE_BELTSTAR_X, MIDDLE_BELTSTAR_Y)
turtle.write('Alnilam')
turtle.goto(RIGHT_BELTSTAR_X, RIGHT_BELTSTAR_Y)
turtle.write('Mintaka')
turtle.goto(LEFT_KNEE_X, LEFT_KNEE_Y)
turtle.write('Saiph')
turtle.goto(RIGHT_KNEE_X, RIGHT_KNEE_Y)
turtle.write('Rigel')

# Draw a line from the left shoulder to left belt star
turtle.goto(LEFT_SHOULDER_X, LEFT_SHOULDER_Y)
turtle.pendown()
turtle.goto(LEFT_BELTSTAR_X, LEFT_BELTSTAR_Y)
turtle.penup()

# Draw a line from the right shoulder to right belt star
turtle.goto(RIGHT_SHOULDER_X, RIGHT_SHOULDER_Y)
turtle.pendown()
turtle.goto(RIGHT_BELTSTAR_X, RIGHT_BELTSTAR_Y)
turtle.penup()

# Draw a line from the left belt star to middle belt star
turtle.goto(LEFT_BELTSTAR_X, LEFT_BELTSTAR_Y)
turtle.pendown()
turtle.goto(MIDDLE_BELTSTAR_X, MIDDLE_BELTSTAR_Y)
turtle.penup()

# Draw a line from the middle belt star to right belt star
turtle.goto(MIDDLE_BELTSTAR_X, MIDDLE_BELTSTAR_Y)
```



```
turtle.pendown()
turtle.goto(RIGHT_BELTSTAR_X, RIGHT_BELTSTAR_Y)
turtle.penup()

# Draw a line from the left belt star to left knee
turtle.goto(LEFT_BELTSTAR_X, LEFT_BELTSTAR_Y)
turtle.pendown()
turtle.goto(LEFT_KNEE_X, LEFT_KNEE_Y)
turtle.penup()

# Draw a line from the right belt star to right knee
turtle.goto(RIGHT_BELTSTAR_X, RIGHT_BELTSTAR_Y)
turtle.pendown()
turtle.goto(RIGHT_KNEE_X, RIGHT_KNEE_Y)

# Keep the window open. (Not necessary with IDLE.)
turtle.done()
```

82 code_02_24_turtle.py

```
# code_02_24_turtle.py
import turtle # Start import module
#turtle.setup(640, 480)
turtle.showturtle() # to display the turtle in its window
turtle.write('Hello World')

# Moving turtle forward
turtle.forward(100) # Moving turtle forward

# Turning the turtle
turtle.left(90) # Turn turtle left by 90 degree
turtle.forward(100) # Moving turtle forward
turtle.right(90) # Turn turtle right by 90 degree
turtle.forward(100) # Moving turtle forward

# Setting the turtle's heading
turtle.setheading(90) # set the turtle's heading to a specific angle
turtle.forward(100)
turtle.setheading(180)
turtle.forward(50)
turtle.setheading(270)
turtle.forward(100)
```

```
# Pen up or down
turtle.forward(50)
turtle.penup()
turtle.forward(25)
turtle.pendown()
turtle.forward(50)
turtle.penup()
turtle.forward(25)
turtle.pendown()
turtle.forward(50)

# Drawing Circles : to draw a circle with a specified radius.
turtle.circle(100) # 100 radius

# Drawing dots
turtle.dot()
turtle.forward(50)
turtle.dot()
turtle.forward(50)
turtle.dot()
turtle.forward(50)


# Changing the pen size and drawing color
turtle.pensize(5)
turtle.pencolor('red')
turtle.circle(100)

# Clear and reset
#turtle.reset()
#turtle.clear()
#turtle.clearscreen()

turtle.reset()

# Moving the turtle to a specific location
turtle.goto(0, 100)
turtle.goto(-100, 0)
turtle.goto(0, 0)

# Filling Shapes
turtle.hideturtle()
turtle.fillcolor('red')
turtle.begin_fill()
```



```
turtle.circle(100)
turtle.end_fill()

# Keep the window open. (Not necessary with IDLE.)
turtle.done()
```

83 code_02_25_inputbox.py

```
# code_02_25_inputbox.py
import turtle
num = turtle.numinput('Input', 'Enter a number', default=10, minval=0, maxval=100)
txt = turtle.textinput('Input', 'Enter your name')
```

84 code_02_26_inputbox_futurevalue.py

```
# code_02_26_inputbox_futurevalue.py
import turtle
turtle.setup(600,600)
#num = turtle.numinput('Input', 'Enter a number', default=10, minval=0, maxval=100)
#txt = turtle.textinput('Input', 'Enter your name')
# Get the desired future value.
#future_value = float(input('Enter the desired future value: '))
# Get the annual interest rate.
#rate = float(input('Enter the annual interest rate: '))
# Get the number of years that the money will appreciate.
#years = int(input('Enter the number of years the money will grow: '))
# Calculate the amount needed to deposit.
#present_value = future_value / (1.0 + rate)**years
# Display the amount needed to deposit.
#print('You will need to deposit this amount:', present_value)

# Inputbox and futurevalue
## Input
name = turtle.textinput('Input', 'Enter your name')
future_value = turtle.numinput(
    'Input',
    'Enter the desired future value: ',
    default=1000,
    minval=0,
```

```
    maxval=1000000000)
rate = turtle.numinput(
    'Input',
    'Enter annual interest rate: (0.01~1.00)',
    default=0.02,
    minval=0.01,
    maxval=1)
year = turtle.numinput(
    'Input',
    'Enter the number of years the money will grow: (1~10)',
    default=1,
    minval=1,
    maxval=10)

## Calculate
present_value = future_value / (1.0 + rate)**year

## Display
turtle.hideturtle()
turtle.penup()
turtle.write('Good afternnon')
turtle.setheading(270)
turtle.forward(10)
turtle.write(name)
turtle.forward(10)
turtle.write('You will need to deposit this amount: ')
turtle.forward(10)
turtle.write(f'{present_value:<10,.2f}')

print('Good afternoon %s' %name)
print('You will need to deposit this amount: %d' %present_value)
print('You will need to deposit this amount: %s' %f'{present_value:<10,.2f}')

turtle.done()
```

